Case Study on Demonstration of Network Communication Models (OSI / TCP-IP) Using a Network Simulation Tool

Aim

To analyze and demonstrate data communication, encapsulation, decapsulation, and packet routing across the OSI and TCP/IP reference models using a network simulation tool.

Objective

- Study different network communication models (OSI 7-layer and TCP/IP 4-layer).
- Analyze the packet flow and encapsulation process in a simulated enterprise network topology.
- Compare the advantages and limitations of various network communication models.
- Demonstrate real-time network traffic workflows including Address Resolution Protocol (ARP), TCP 3-Way Handshake, and HTTP GET request workflows.

Selected Network Simulation Tool & Scenario

Selected Simulation Tool

Cisco Packet Tracer v8.x

Overview

A simulated network environment allows network administrators and engineers to:

- Build multi-subnet network topologies.
- Inspect layer-by-layer Protocol Data Units (PDUs) in real time.
- Observe Address Resolution Protocol (ARP) execution.
- Trace TCP 3-Way Handshake connection setup.
- Analyze HTTP client-server data exchange and decapsulation.

Topology Setup & Components

Client Host Module

- **PC-A:** IP 192.168.1.10 /24, Default Gateway: 192.168.1.1, MAC: 0060.2F11.AAAA

Switching Module

- **Switch S1:** Cisco Catalyst 2960 (Layer 2 Switching)

Routing Module

- **Router R1:** Cisco 1941 ISR (LAN Interface: 192.168.1.1/24, WAN Interface: 10.0.0.1/24)

Server Module

- **Server-A:** IP 10.0.0.50 /24, Default Gateway: 10.0.0.1, MAC: 00D0.BA22.BBBB (HTTP Port 80 Active)

Network Communication Models Analysis

Network Model	Suitability	Advantages	Limitations
OSI Model (7 Layers)	Medium	Comprehensive theoretical framework; clearly separates service, interface, and protocol concepts.	Complex implementation; redundant session/presentation layers; not universally deployed in commercial internet.
TCP/IP Model (4 Layers)	Very High	Pragmatic, highly scalable; universal internet standard; lightweight header overhead; simple 4-layer structure.	Does not clearly distinguish between service, interface, and protocol concepts.

Why TCP/IP is Best for Real-World Implementation?

Reasons

- Universal commercial adoption across all modern operating systems and networking hardware.
- Streamlined 4-layer architecture minimizes protocol processing latency.
- Seamless integration with underlying physical link-layer technologies (Ethernet, Wi-Fi, 5G).
- Robust routing capabilities natively supported by modern internet infrastructure.

Data Encapsulation & Transmission Process

```
Application Data (HTTP Request)
        ↓
Transport Layer Encapsulation (TCP Segment Header Added)
        ↓
Internet Layer Encapsulation (IP Packet Header Added)
        ↓
Network Access Layer Encapsulation (Ethernet Frame Header & Trailer Added)
        ↓
Physical Bit Transmission (Raw Binary Signals over Media)
        ↓
Decapsulation at Destination (Server-A Processes HTTP Request)
```

Simulation Execution & Analysis Phases

1. Topology Configuration & IP Addressing

Setup network topology in Cisco Packet Tracer with PC-A, Switch S1, Router R1, and Server-A.

- **Deliverables:** Network Topology Diagram, IP Addressing Scheme.

2. Address Resolution Protocol (ARP) Discovery

PC-A initiates communication with Server-A (10.0.0.50). Recognizes destination is outside local subnet and broadcasts ARP Request for Gateway 192.168.1.1.

- **Deliverables:** ARP Request/Reply PDU Captures.

3. Connection Setup: TCP 3-Way Handshake

PC-A establishes a reliable connection with Server-A on Port 80.

- **SYN:** PC-A sends SYN Segment (Seq=0).
- **SYN-ACK:** Server-A replies with SYN-ACK (Seq=0, Ack=1).
- **ACK:** PC-A sends ACK Segment (Seq=1, Ack=1). Session established.

4. HTTP Web Data Request & Decapsulation

PC-A issues an HTTP GET /index.html request formatted at Layer 7.

PDU Inspection Breakdown:

OSI / TCP-IP Layer	PDU Header Component & Value
Application Layer	HTTP Payload: GET /index.html HTTP/1.1
Transport Layer	TCP Header: Src Port: 49152, Dst Port: 80, Seq: 1, Ack: 1
Internet Layer	IPv4 Header: Src IP: 192.168.1.10, Dst IP: 10.0.0.50, Protocol: 6 (TCP)
Network Access Layer	Ethernet Frame: Src MAC: 0060.2F11.AAAA, Dst MAC: 0002.4A11.1111 (Gateway R1)

5. Routing & Hop-by-Hop Frame Rewriting

Router R1 receives the frame, strips Layer 2 header, inspects Layer 3 IP header, and re-encapsulates the IP packet into a new Ethernet frame with Destination MAC = Server-A (00D0.BA22.BBBB).

6. Decapsulation & Response Delivery

Server-A receives the raw physical bits, decapsulates bottom-up through Network Access, Internet, and Transport layers, delivering the HTTP payload to the Web Server application.

Comparison of Communication Models

Criteria	OSI Model	TCP/IP Model
Layer Count	7 Layers	4 Layers
Approach	Theoretical / Conceptual	Practical / Implementation
Session & Presentation Layers	Explicitly defined as separate layers	Merged into Application Layer
Protocol Dependency	Protocol Independent	Protocol Dependent (TCP/IP Suite)
Header Processing Overhead	Higher (due to extra layering)	Lower (optimized for speed)
Suitable for Practical Networks	No (Used for teaching/reference)	Best (Universal internet standard)

Results

- The simulated web transaction successfully demonstrated multi-layer encapsulation and decapsulation across subnet boundaries.

- TCP/IP 4-layer model provided efficient data delivery with minimal processing overhead.
- Intermediate routers processed packets up to Layer 3 (IP header), changing MAC addresses hop-by-hop while maintaining end-to-end IP addresses.
- Therefore, the TCP/IP model is the most practical and efficient standard for modern network communication.

Conclusion

The case study demonstrates that while the OSI 7-layer model provides an essential theoretical foundation for understanding network modularity, the TCP/IP 4-layer model serves as the practical, real-world framework driving modern internet communications. Network simulation tools like Cisco Packet Tracer effectively visualize layer-by-layer PDU encapsulation, enabling deep operational understanding of network protocol interactions.